

Alpecca Feature And Function Skeleton

Corrected source-backed status map. Red means unsafe to activate; green requires runtime and security evidence.

STATUS	DEFINITION
DONE	Live, tested, runtime-verified, documented, and not security-blocked
PARTIAL	Useful implementation exists; a required gate remains open
BLOCKED	Unsafe to activate until remediation passes
NOT STARTED	No production implementation
PARKED	Intentionally deferred experiment
SUPERSEDED	Historical claim replaced by current evidence

LANE	ROLE	CAPACITY / GUARANTEE
Local laptop	Authoritative CoreMind, state, policy, identity, approvals, live fallback	Approximately 24 GB DDR5-4800; RTX 3050 Laptop GPU with 4 GB VRAM
Hugging Face ZeroGPU	Optional stateless deep/vision/texture/batch inference	Ephemeral, quota-governed, runtime-probed; never counted as local capacity
Google notebook / Colab	Optional stateless accelerated inference or batch jobs	Ephemeral; GPU, RAM, uptime, and limits are not guaranteed

Feature Skeleton: Tiers 1-5

TIER	FEATURE	STATUS	CURRENT TRUTH
Tier 1 - Foundation runtime	FastAPI, WebSocket, runtime status	PARTIAL	Works; identity, turn isolation, and cancellation barriers remain
	Ollama routing and offline fallback	PARTIAL	Local path works; cloud privacy classification is incomplete
	SQLite state, memory, journal, proposals	PARTIAL	Functional; backup/concurrency/transaction hardening remains
	Remote auth and tunnels	BLOCKED	Exposed root token plus HTML self-authentication bypass
	Singleton and active portal	NOT STARTED	No OS mutex, boot identity, lease, or epoch fencing
Tier 2 - Cognition and agency	Soul seven-subagent arbitration	DONE	Implemented without adding or bypassing subagents
	CoreMind response loop	PARTIAL	Global speaker/history can leak across surfaces
	Constrained choice points	DONE	Living, Soul tie-break, and proactive judge with fallback
	Tool registry and planner	PARTIAL	Bounded; guest capability and approval identity are unsafe
	Cue envelope and commitment ledger	NOT STARTED	No durable correction/confirmation/action closure contract
Tier 3 - Memory and Mindpage	Keyword/FTS recall and backfill	PARTIAL	Useful; semantic scoring can admit unrelated memories
	Mindpage Layer A	PARTIAL	Ledger/writeback/faults work; overflow and buried indexing remain
	Conversation/privacy partitioning	BLOCKED	Short-term context is shared across app/guest/Discord/creator
	Resource pressure sensing	PARTIAL	Context grounded; whole-machine draft is unwired
	Approved pagefile broker	BLOCKED	Draft math, proof, recheck, and verification are unsafe
	llama.cpp KV persistence	PARKED	Downloaded experiment; not integrated
Tier 4 - Autonomy and improvement	Proactive/living behavior	PARTIAL	Multiple outward loops lack one initiative budget
	Recursive self-improvement	PARTIAL	Tunables exist; consumer wiring and trials are weak
	External action approval	BLOCKED	Caller booleans/global confirmation are not identity-bound
	Action receipts and promise closure	NOT STARTED	No durable evidence contract
Tier 5 - Automation	Routines and watchers	PARTIAL	Empty/off by default; deletion and unified scheduling remain
	Background job coordination	PARTIAL	Timeouts do not cancel worker threads
	MCP federation	PARKED	Deferred external surface

Feature Skeleton: Tiers 6-9

TIER	FEATURE	STATUS	CURRENT TRUTH
Tier 6 - Experience and embodiment	House HQ and virtual app	PARTIAL	Working surfaces; portal ownership and QA remain
	V4 VRM body and physics	PARTIAL	74 joints; scale/grounding/collider/motion gates remain
	Expression and gesture control	PARTIAL	Expression latch and looping VRMA issues remain
	Locked design QA	PARTIAL	Hair/clip/readability and turntable evidence remain
Tier 7 - Voice, senses, and communication	Local TTS voice	PARTIAL	Kokoro/F5 work; queue/resource coordination remains
	Image and file perception	PARTIAL	Scope/privacy/MIME/cloud consent remain
	Audio perception	NOT STARTED	Discord audio and live receive/transcription absent
	Discord autonomy	BLOCKED	Guest tools, context leakage, spam, approval gaps
	Creator contact outbox	NOT STARTED	WIP skeleton only; no routes/workers/tests
Tier 8 - Cloud and continuity	Computer use	BLOCKED	Unsafe under current remote auth and global confirmation
	Hugging Face art/runtime assets	PARTIAL	Correct storage lane; publish/availability remain operational
	ZeroGPU / Google notebook	PARTIAL	Optional ephemeral adapters; privacy policy incomplete
	Cloudflare R2 shell	BLOCKED	Rebuild after token rotation and bundle cleanup
	Mindscape continuity	BLOCKED	Fail-open/auth/restore defects
Tier 9 - Governance	Cloud egress broker	NOT STARTED	No deny-by-default data/provider policy
	Design lock and canonical docs	DONE	Current hierarchy and locked character design exist
	Creator identity and secret lifecycle	BLOCKED	No authoritative principal or scoped secret lifecycle
	Grounded affect/self-report	PARTIAL	State is real; some prompts overclaim inner experience
	Only pagefile as planned OS mutation	PARTIAL	Policy exists; broker is not safe or active

Critical implementation order

#	PHASE	STATUS	BUILD	EXIT GATE
0	Truth baseline and freeze	DONE	Corrected docs, local/cloud labels, WIP boundaries, status rules, and encrypted recovery baseline.	Diagrams match evidence; encrypted DB/V4/VRoid restore verified; 34 GB/H100 local claim removed.
1	Emergency security containment	BLOCKED	Disable tunnels/computer control; preserve Alpecca's public identity value; stop using it for bearer auth; add a separate protected authorization secret; fix HTML auth.	Anonymous HTTP/WS executes nothing; no authorization secret in bundles; public identity grants no privilege.
2	Creator identity, singleton, portal lease	NOT STARTED	Server-derived principal, pairing/sessions, bridge credentials, Windows mutex, signed health, lease epoch.	One CoreMind and one writer; spoofed creator claims and stale portals fail.
3	Turn transactions and context isolation	NOT STARTED	Immutable turn/conversation/actor/surface/privacy context, cancellation, commit barrier, scoped retrieval.	No cross-actor leak; timed-out workers cause zero late effects.
4	Cue, commitment, action closure	NOT STARTED	Structured cues plus durable commitments and receipts; completion language requires evidence.	Every promise executes, awaits approval, fails, or cancels honestly.